

Discussion of Understanding Artisanal Mining

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APE Annual Meeting – July 22, 2026

- **Rare access:** an original survey of 1,229 artisanal gold miners in Madre de Dios.
- **A sharp puzzle:** 74% rank illegal mining as most profitable, yet roughly three quarters want to formalize and use clean technology.
- **Two empirical exercises:** a latent-class segmentation and a randomized conjoint over policy-support packages.

A valuable diagnostic of heterogeneity and stated support demand; the key question is how far these measures take us toward actual upgrading.

Comment 1: Three classes vs two?

- **Ready vs. constrained:** intentions +0.23 SD, WTP +0.68 SD, and donations +0.48 SD.
- **Capable vs. constrained:** -0.05, +0.10, and +0.08 SD; all confidence intervals include zero.
- The clearest result is a standout ready group in terms of behavioral responses.
- I would also bootstrap the whole classification + regression to account for post-selection to classes.

The evidence may support a simpler targeting story more strongly than a three-step readiness ladder.

Comment 2: Who actually decides?

- The paper calls 267 owners **decision-makers**; the other 962 respondents are workers.
- Equipment, credit, certification, and buyer contracts are operation-level choices; intentions and conjoint responses are individual-level measures.
- Since 81% of the ready group are owners, versus 7% of the capable group, the classes may mix readiness with authority.
 - ✓ Show owner-only and operation-level results.

The policy target is an operation with authority to invest, not simply a respondent with positive intentions.

Comment 3: Offers are not constraints

- The AMCE identifies how a randomized feature changes **hypothetical profile selection**, and not actual adoption or constraint bindingness.
- Regulation (60%) and training (53%) are reported more often as barriers than finance (19%), although interest-free credit has one of the largest AMCEs.
- The offers differ in generosity and salience; positive interactions support non-additivity in stated choice, not yet production complementarity.

Perhaps it would be good to frame this as demand for support features, then test equal-cost instruments and bundles using observed behavior.

Comment 4: Upgrading is a sequence, not an intention

- The paper seems to combine three upgrading margins: adopting the gravity table, formalizing the operation, and reaching certified clean-gold markets.
- The survey reveals an **aspiration–investment gap**: 75% want to use clean technology, but only 24% previously knew the gravity table and knowledgeable respondents' WTP peaks near PEN 15,000 against a market price near PEN 40,000.
- This gap does not isolate one constraint. It could reflect uncertain returns, limited credit, missing complementary inputs, or the fact that many respondents cannot authorize the investment.
- Verhoogen's taxonomy clarifies the mechanisms: **know-how** about the technology, **input access** to finance and equipment, and **output demand** from buyers willing to reward clean gold.

- **Contribution:** rare access, a memorable aspiration–profitability puzzle, and a creative policy-design exercise.
- **Segmentation:** the ready group clearly separates; establish whether the capable group is distinct and carry classification uncertainty through inference.
- **Decision unit:** distinguish willingness from authority and move toward owner- and operation-level outcomes.
- **Estimand:** the conjoint identifies stated demand for offers, not binding constraints, realized adoption, or production complementarity.

Appendix: A mechanism-based classification of the 12 features

Output demand and contracting

- Clean-gold price premium
- Five-year buyer price guarantee
- Annual buyer fair / market access
- Certification-cost support (partly)

Input supply and cost

- Shaker-table discount
- Interest-free credit
- Maintenance coverage
- Secure logistics / safehouse

Know-how

- Shaker-table training
- Export and logistics training
- Formalization training
- Certification training

Important constraints outside the conjoint

- Regulation and enforcement
- Land tenure
- Security

Appendix: Experiment identifies profile choice, not adoption or constraints

Profile A

Table: 50% discount
Buyer: guarantee
Training: free
6 of 12 shown

Profile B

Table: market price
Buyer: no guarantee
Training: none
6 of 12 shown

Neither

One task = one choice
among A, B, and neither

- **Tasks:** $1,229 \times 5 = 6,145$ decision occasions.
- **Profiles:** A and B become two alternative-level rows, giving 12,290 profile observations.
- The AMCE estimates how a randomized profile feature changes the probability that the profile is selected.
- Standard errors are clustered by respondent because tasks repeat; the estimand is still stated choice.